

PROGRAM 5

PSEUDOCODE

ALGORITMA PENENTUAN CLUSTER

DEKLARASI:

```
A = (2, 1, 3)      : real // Pusat Cluster A
B = (1, -4, 6)     : real // Pusat Cluster B
C = (-2, 3, -2)    : real // Pusat Cluster C
x1, x2, x3         : real // Koordinat titik U
dA, dB, dC         : real // Jarak Euclidean ke cluster
Cluster           : string // Hasil akhir kategori cluster
```

START

```
INPUT x1, x2, x3
```

```
dA ←  $\text{SQRT}((x1 - 2)^2 + (x2 - 1)^2 + (x3 - 3)^2)$ 
```

```
dB ←  $\text{SQRT}((x1 - 1)^2 + (x2 + 4)^2 + (x3 - 6)^2)$ 
```

```
dc ←  $\text{SQRT}((x1 + 2)^2 + (x2 - 3)^2 + (x3 + 2)^2)$ 
```

```
IF (dA ≤ dB) AND (dA ≤ dc) THEN
```

```
    Cluster ← "Cluster A"
```

```
ELSE IF (dB ≤ dA) AND (dB ≤ dc) THEN
```

```
    Cluster ← "Cluster B"
```

```
ELSE
```

```
    Cluster ← "Cluster C"
```

```
END IF
```

```
OUTPUT "Titik U termasuk ", Cluster
```

```
END
```

FLOWCHART

